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Revisiting Mercury Wetting Properties and their Implications for Pore Throat and Pore Body Analysis Using Mercury Injection Capillary Pressure in Porous Media Characterization

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Abstract

The wetting characteristics of mercury, specifically contact angle and surface tension (or interfacial tension, IFT), are required parameters, for the conversion of MICP (mercury injection capillary pressure, P_c) data into pore-throat sizes, P_c curves of reservoir systems, and pore body-to-throat ratio. The current study investigates the effect of lithology and surface preparation on mercury contact angle and IFT measurements in air with the objective of enhancing MICP based porous media pore structure characterization, including pore typing and pore body-to-throat ratio assessment.

In this work, mercury contact angle and IFT (between mercury and air) were measured using an optical pendant drop tensiometer on a variety of lithological surfaces including limestones, dolostones, coarse-grained and fine-grained sandstones, and clay-rich sands. The contact angle was measured on three different areas of a prepared rock surface, in both preserved rough surface and diamond-polished smooth surface, to assess effect of surface roughness on contact angle measurement. The tests were done under static conditions to minimize the influence of surface pores and the possibility of mercury drops falling into pore cavities. As such, the mercury drop was observed for a short period of time; only five seconds. The optical tensiometer captured image was then analyzed with an automated drop-profile fitting method to determine contact angle, and in the process, IFT was also obtained by analyzing the shape of the pendant drop.

The measured contact angles at the three different locations on a rock surface were close to each other for the limestone and dolostone samples, indicating homogeneity of their surfaces. However, discrepancies between measurements on the three different locations were observed for the sandstone and clay-rich sand samples. XRD (X-ray diffraction) tests showed that the limestone and dolostone were over 98% calcite and 99% dolomite, respectively. However, the sandstone and clay-rich sand showed a richer variety of mineralogy content, which suggests that heterogeneity in surface mineralogy may result in heterogeneity in the contact angle measurement. These discrepancies can also be explained by the presence of pore cavities on the rock surfaces, which could make the mercury drop unstable and the measurement less accurate.

From this study, the average mercury contact angle measured was 138°, only slightly less than the commonly used value of 140°, while average IFT was 417 dynes/cm, 14% less than the often defaulted

485 dynes/cm. The combined effect resulted in an average 17% reduction in pore throat size derived from MICP intrusion and pore body size from MICP extrusion measurement, with little effect on the derived pore body-to-throat ratio.

Introduction

Although mercury, also named hydrargyrum (Hg), is considered a hazardous material, because of its unique characteristics of being heavy with an atomic number of 80 and a density of 13.546 g/cc, and existing as a liquid at ambient temperature and pressure, it is widely used in the industry and society. In petrophysics, the use of mercury to measure capillary pressure (P_c) of reservoir rocks, i.e., the mercury injection capillary pressure (MICP) curve, is a routine laboratory test and the most commonly used method to derive reservoir P_c , compared to other routinely used methods, the centrifuge and porous plate methods.

To process and interpret data such as MICP, air (vacuum)-mercury interfacial tension (IFT, or σ) and air (vacuum)-mercury-solids contact angle (θ) are required, and textbook values of $\sigma = 485$ dynes/cm and $\theta = 140^\circ$ are almost always defaulted, universally, though the needs to evaluate their representativeness has been noted (Ma et al., 1991) with the below cited concerns:

- Effect of contact angle hysteresis (Morrow, 1990; Ma et al., 1996)
- Effect of pore surface characteristics (Morrow, 1975; Sauerer et al., 2020; Ma et al., 2021; Singer et al., 2023 and 2024).
- Effect of pore shape (Ma et al., 1996; Ma et al., 2023; Ma et al., 2024).

In this study, the effect of rock types, lithology, and surface preparation on mercury contact angle measurements is investigated and the mercury IFT in air is measured with the objective of reducing their uncertainties in petrophysical applications to enhance MICP based pore structure characterization, including pore typing and pore body-to-throat ratio assessment. The contact angle on a variety of lithological surfaces, as well as a glass slide (smooth surface), and the IFT in air are measured using optical pendant drop tensiometer equipment. The effect of the surface preparation, preserved surface versus diamond-polished, on the mercury contact angle is also investigated. The optical tensiometer was used to record and analyze images of pendant and sessile drop, allowing for the determination of contact angle and IFT.

Literature Review

Mercury contact angle

The wetting characteristics of mercury, specifically contact angle (θ) and surface tension (or interfacial tension, IFT, σ), are required parameters for the conversion of the measured mercury injection capillary pressure (MICP, or P_c) data into reservoir properties of

- pore-throat sizes,
- reservoir system (such as oil and water) P_c curves, and
- MICP based pore body-to-throat relationships if MICP intrusion and extrusion data is available.

To derive pore throat size from P_c curves (Pickell et al., 1966; Swanson, 1977; Ma et al., 1991; Ma et al., 1996), it is often assumed that the porous sample is made of small cylindrical capillaries, P_c is the pressure differences between the non-wetting phase mercury and wetting phase air (vacuum) which is required to overcome in order for mercury to penetrate a pore throat of diameter d , as given by the Washburn equation:

$$d_p = \frac{4\sigma\cos\theta_w}{P_c} = \frac{4\sigma(-\cos\theta_{mw})}{P_c} \quad (1)$$

where σ is the IFT (dynes/cm), θ the contact angle (degrees) and d_p the pore throat diameter (μm). Subscripts w and nw represent wetting phase and non-wetting phase.

The contact angle between mercury and the porous sample used in most Pc calculation is normally 140° regardless of the material under investigation. From Eq. 1, it is seen that if the contact angle is, for example, 20° different from the often-defaulted value of 140° , the error of the calculated pore size would be 35%, illustrating the importance of the accuracy of the contact angle used in Pc based pore typing.

Values of contact angle can be measured by static or dynamic methods. The static method, or sessile drop method, uses an optical microscope and goniometer to quantify the non-wetting behavior of mercury droplet on a solid surface (Sauerer et al., 2020). The dynamic method measures the pressure (equivalent to Pc) needed to force a mercury droplet of known volume through a capillary tube with known diameter (d_p) and applies the Washburn equation (Eq. 1) to determine the contact angle θ , with a given IFT (which is often assumed):

$$\cos\theta_{nw} = -\frac{P_c d_p}{4\sigma} \quad (2)$$

The dynamic method is a better approximation for the actual intrusion conditions, including evacuating the pores prior to intrusion, while the static method measures the equilibrium contact angle. Apparently, because it is dynamic, the effect of hysteresis should be considered. In addition, the measurements of the dynamic method are more difficult to control, which may result in higher measurement uncertainties. Fabricating perfect circular cylindrical tubes can also be challenging, so values of d_p may also carry larger uncertainty.

Groen et al. (2002) applied both static and dynamic methods to measure the contact angle for various materials and both methods gave similar results, while important deviations from 140° were observed in carbon, $\theta_{\text{carbon}} \geq 150^\circ$ and cement-like materials, $\theta_{\text{cement}} \leq 130^\circ$. Ethington (1990) measured the contact angle by the static method on pure mineral surfaces that were smoothed to have zero porosity effect and found that $\theta_{\text{quartz}} = 136^\circ \pm 3^\circ$, $\theta_{\text{calcite}} = 136^\circ \pm 3^\circ$, and $\theta_{\text{biotite}} = 138^\circ \pm 4^\circ$, all slightly lower than the default value of 140° , but the measurement on the clay sample was significantly higher at $\theta_{\text{clay}} = 158^\circ \pm 6^\circ$, which is probably the highest mercury contact angle reported in the literature.

The sessile contact angle is most representative of the intrinsic value; however, it has the uncertainty as to whether the measured angle is the advancing (θ_{adv}) or receding angle (θ_{rec}). A more advanced test capturing the angle on an incline would allow for quantification of the difference between advancing and receding angles. The test may not be difficult for oil/water systems, but challenging for mercury, due to its high density that the mercury drop would roll very fast. The receding angle, which is the contact angle between a fluid and a solid when the fluid is receding or withdrawing is always smaller than or equal to the advancing angle which occurs during the fluid advancing or spreading on the surface ($\theta_{\text{adv}} \geq \theta_{\text{rec}}$). The difference between θ_{adv} and θ_{rec} is defined as contact angle hysteresis, which is greater than or equal to zero, $\Delta\theta = \theta_{\text{adv}} - \theta_{\text{rec}} \geq 0$, depending on factors such as surface roughness and chemical heterogeneities. The rougher the surface, the higher the contact angle hysteresis, as Marmur stated that the contact angle hysteresis can vary from a few degrees to over 100 (Marmur et al., 2017). For a strongly non-wetting liquid like mercury, $\Delta\theta$ is expected to be small, as observed by Ueda (2015).

Ellison et al. (1967) measured an equilibrium contact angle value of 134° ; they measured a maximum advancing angle of 147° and minimum receding of 122° , both reached the similar equilibrium value in 10-30 seconds. Smithwick (1988) measured the contact angles of microscopic mercury droplets under vacuum. Immediately after droplets formed (by condensation), values from a maximum 153° and a minimum 113° were measured. After equilibration for 2 weeks, a constant contact angle of about 130° was obtained. Compared to the work by Ellison et al. (1967), a wider range of contact angles was initially observed, and the equilibration time for the small droplets was thousands of times longer. The longer equilibration time could be because the smaller drops (0.2 mm in diameter) are small in mass that they are influenced significantly

less by background sources of low-frequency vibration. Table 1 summarizes values of the above mercury contact angle measurements, in air and at room temperature, in the three decades from 1960s (Ellison et al., 1967) to 2000s (Groen et al., 2002).

Table 1—Summary of measured mercury contact angles at room temperature.

Solids	Condition	Contact Angles θ (deg)	Methods	Reference
Glass		134° , $\theta_{adv}=147^\circ$ / $\theta_{rec}=122^\circ$	Static	Ellison et al. (1967)
Nickel	Polished	139° , $\theta_{adv}=148^\circ$ / $\theta_{rec}=123^\circ$	Static	Ellison et al. (1967)
Quartz	Polished	133° , $\theta_{adv}=147^\circ$ / $\theta_{rec}=115^\circ$	Static	Ellison et al. (1967)
Stainless steel	Polished	133° , $\theta_{adv}=146^\circ$ / $\theta_{rec}=124^\circ$	Static	Ellison et al. (1967)
Teflon	Polished	133° , $\theta_{adv}=157^\circ$ / $\theta_{rec}=104^\circ$	Static	Ellison et al. (1967)
Tungsten	Polished	131° , $\theta_{adv}=142^\circ$ / $\theta_{rec}=121^\circ$	Static	Ellison et al. (1967)
Glass		$130^\circ \pm 2^\circ$	Static	Smithwick (1988)
Biotite	Polished	$138^\circ \pm 4^\circ$	Static	Ethington (1990)
Calcite	Polished	$136^\circ \pm 3^\circ$	Static	Ethington (1990)
Clay	Polished	$158^\circ \pm 6^\circ$	Static	Ethington (1990)
Quartz	Polished	$136^\circ \pm 3^\circ$	Static	Ethington (1990)
Carbon	Compressed powder	$\geq 150^\circ$	Static & dynamic	Groen et al. (2002)
Cement	Compressed powder	$\geq 130^\circ$	Static & dynamic	Groen et al. (2002)
Oxidic materials	Compressed powder	$\sim 140^\circ$	Static & dynamic	Groen et al. (2002)
Glass		$136^\circ \pm 1^\circ$	Static	This study
Limestone	Rough	$137^\circ \pm 1^\circ$	Static	This study
Limestone	Polished	$134^\circ \pm 2^\circ$	Static	This study
Dolostone	Rough	$134^\circ \pm 1^\circ$	Static	This study
Dolostone	Polished	$141^\circ \pm 1^\circ$	Static	This study
Coarse-grained sandstone	Rough	$137^\circ \pm 1^\circ$	Static	This study
Coarse-grained sandstone	Polished	$135^\circ \pm 3^\circ$	Static	This study
Fine-grained sandstone	Rough	$139^\circ \pm 8^\circ$	Static	This study
Fine-grained sandstone	Polished	$149^\circ \pm 1^\circ$	Static	This study
Shaly sand	Rough	$140^\circ \pm 6^\circ$	Static	This study
Shaly sand	Polished	$138^\circ \pm 2^\circ$	Static	This study
Total average		137.6°		
Average sedimentary rocks		139.1°		
Ave sedimentary clean rocks		137.6°		
Average of this study results		138.4°		

Note that mercury cleanliness or contamination could also affect the measurements of contact angle, though Cook and Hover (1991) found that the contact angle between mercury and hardened cement paste was not significantly affected by the cleanliness of the mercury. Further study is required to determine if this applies also to sedimentary rock.

Mercury interfacial tension IFT

Besides contact angle, the other factor influencing the P_c curve is the interfacial tension (IFT, σ). The earliest value recorded of mercury IFT was of 563 dynes/cm by Morveau (1773). A detailed study of existing literature up to the early 1970's (Wilkinson, 1972) shows a wide range of values for the mercury IFT determined by a wide range of techniques. The reported values do not only vary with the method used but also between different measurements using the same technique. Mercury has high IFT, and impurities would be expected to interact strongly at the surface. The presence of external impurities (i.e., gas phase) may explain some of the low values, but it is unlikely that all the low measured values are due to this factor alone. Figure 1 shows the values recorded for the mercury IFT since 1805 by different methods (left) and their distribution (right).

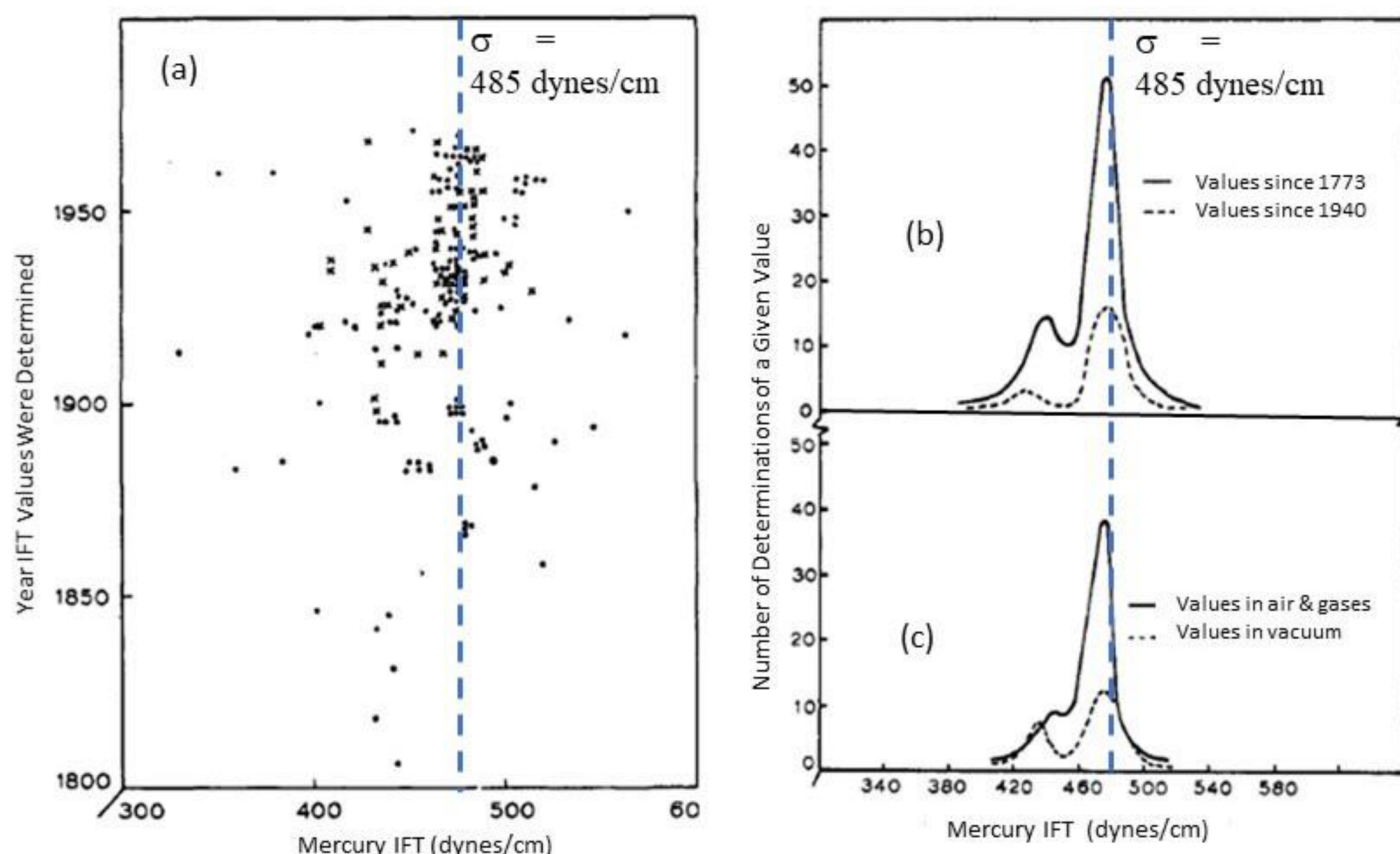


Figure 1—Mercury IFT values recorded since 1805 and their statistical distributions (from Wilkinson, 1972).

The majority peak of the IFT distribution may be narrowed with time, but the range of values did not change much (Figure 1b). The sessile drop technique has given the biggest range (360-560 dynes/cm). While the measurement techniques have improved with time, extreme low and high values are still being recorded (Figure 1b). The existence of the two separate peaks lead to the conclusion that a mercury surface atom may have different states, one corresponding to a stable state with a low IFT and the other to a state with a high IFT (Gmelin, 1960). Values recorded since 1940 still show two peaks, but the short one is reduced. In Figure 1c, values recorded in vacuum are plotted separately from those in air and other gases which indicates that the lower peak is mainly due to the values obtained in vacuum, which may be related to the conventional MICP measurements that a vacuum is applied at the beginning of the test. The wide range in IFT could also be due to external contaminants, e.g., water vapor, organic materials, loose solid particles, internal impurities (dissolved metals) and poor experimental technique, however they are probably not the main reasons, though normal atmospheric gases (nitrogen, oxygen, carbon dioxide, carbon monoxide, hydrogen, and the rare gases) may not affect mercury IFT (Wilkinson, 1972).

The most used method for measuring IFT in the petroleum industry is the pendant drop method, described firstly by Bashforth and Adams (1883). The technique is based on extracting a drop shape from the Young-Laplace equation:

$$\sigma = -\Delta\rho g \frac{z}{\kappa} \quad (3)$$

where $\Delta\rho$ is the density difference between the drop phase and the surrounding, g is gravitational acceleration, z is the vertical distance from the drop apex, and κ is the drop curvature.

The pressure difference between the drop and the surrounding phase creates a force that tries to minimize the surface area. The negative sign signifies that the direction of the normal vector is pointing out of the liquid surface, so pressure increases when going from air to liquid, i.e., pressure increases when going opposite the

normal vector. The drop shape extracted from Eq. (3) is fitted to a measured drop shape extracted from an image. The quality of the measurement depends on the quality of the image of the fluid drop. The precision of the measurement is determined by the drop volume. The Worthington number, Wo , which scales from 0 to 1, is used to account for the volume effect. Larger values of Wo mean greater precision. Drops that are closer to the critical detachment volume offer the most accurate and precise measurements (Berry et al., 2015) because it is at this moment that the interfacial force is just balanced by the gravitational force.

The main assumptions in determining IFT using Eq. 3 are that

- the drop is symmetric about a central vertical axis, and
- the drop is not in motion, i.e., IFT and gravity are the only forces shaping the drop.

Due to the high density of the mercury, the hanging drop must be very small to prevent it from quickly falling. On the other hand, for smaller drop sizes the image resolution would be relatively low to facilitate the drop shape analysis. There are other experimental methods for measuring IFT, like for example the Du Nouy Ring and Wilhelmy Plate, however they are not suitable for mercury since for these methods to work the fluid must be wetting.

In practice, default values of mercury contact angle of 140° and IFT of 485 dynes/cm are often used with limited information about factors that may have various influence on them, like lithology, surface preparation and mercury cleanliness. This is partly because mercury is an HSE hazardous material that many laboratories are unable or unwilling to deal with it, except for special tests in closed systems such as MICP. In addition, it is challenging to use Eq. 3 for mercury IFT measurement because it is shiny and difficult to image, thus specialized equipment is required. Also, due to its high density it is difficult to obtain a full-bodied drop needed to perform the IFT calculations. If the needle is too large, the pendant drop will fall, while if it is too small, the small drop image resolution may be too low for accurate image analysis. Smaller drop is most likely the main source of the measurement error.

Methodology

As stated above, mercury is an HSE hazardous material, thus few laboratories in the industry are equipped to deal with it for special studies, such as measuring IFT and contact angles. In this study, a specialized commercial laboratory is contracted to conduct a series of customized tests using mercury. The methods used are the optical tensiometer measurements to measure the contact angle of mercury on sample surface as a function of time, and IFT of mercury in air, MICP drainage and imbibition curves are measured for characterizing the pore throats and body sizes. For determining the mineralogy and clay specifications, XRD tests are performed, though it has been reported that FTIR may have advantages over XRD for detecting trace amount minerals (Al-Hamad et al., 2025).

In this study, the optical tensiometer sessile drop method with mercury is used on the following six types of solid surfaces to measure the contact angle:

1. Glass slide (smooth surface)
2. Sandstone fine-grained
3. Sandstone coarse-grained
4. Sandstone shaly
5. Limestone
6. Dolostone

Dual measurements of contact angle were performed on each sample, after initial cleaning using a Soxhlet extraction process with toluene and methanol (Singer et al., 2022; Al Qatari et al., 2024):

1. the first set of measurements on cleaned sample material with surface roughness preserved, and
2. the second set on cleaned sample material on a diamond-polished surface.

In each case

1. three measurements were done,
2. with three mercury drops
3. at three different locations on the rock surface.

The pendant drop shape analysis was used to measure mercury IFT in air (Eq. 3). Typically, this is performed in conjunction with (prior to) setting the drop upon the sample surface for the contact angle. However, in this work, we had to go back to measure the pendant drop IFT with an appropriate needle gauge because the initial drop sizes were not large enough to accurately analyze the drop shape.

XRD mineralogy and MICP (drainage and imbibition) and were performed on each sample to associate the textural and mineralogical controls with the various wetting properties. The surface influenced contact angle and IFT values were incorporated into the MICP pore throat/body size calculations to compare against the widely assumed default values of 140° contact angle and 485 dynes/cm IFT.

Below is a generalized step-by-step workflow for this study:

1. Core samples were mechanically split using a splitting wedge to expose surface with native surface roughness preserved. Each sample was individually placed underneath an optical pendant drop tensiometer.
2. Mercury was hung from a narrow needle on the pendant drop tensiometer. The drop shape was analyzed optically to determine IFT related properties.
3. The mercury was dropped onto the surface of the testing sample, from which the drop shape on the sample surface was analyzed optically to determine contact angle.
4. Mercury was removed from the sample surface, and the surface was polished using a diamond-tipped grinding wheel. Each sample was individually placed underneath an optical pendant drop tensiometer.
5. Steps 2 and 3 were repeated with the diamond polished samples.
6. The remaining sample material was cleaned using a Soxhlet extraction process with toluene and methanol.
7. The post-extraction material from each sample was split for MICP and XRD measurements to help to interpret the contact angle data from the above optical pendant drop measurements and evaluate their impacts.

Sample Selection

The selected rock samples for testing are limestone, dolostone, coarse-grained sandstone, fine-grained sandstone, and clay rich shaly sandstone, as listed in Table 2. Estimates of clay bound water (CBW), attributed to the NMR T_2 distribution components below 3 ms cutoff are also included in Table 2. Optical microscope images of the samples are shown in Figure 2.

Table 2—Selected samples and their respective NMR porosity, Swanson air permeability and NMR clay bound water content.

Sample type	Rock sample	ϕ (p.u.)	k_{air} (mD)	CBW using 3 ms T_2 cutoff (% PV)
Limestone	Indiana (ILS)	16.04	55.4	~ 0
Dolostone	Silurian dolomite (SD)	12.24	74.4	~ 0
Coarse-grained sandstone	Berea (BER)	19.71	144	3
Fine-grained sandstone	Kentucky (KEN)	15.6	1.28	17
Clay rich shaly sand	Subclarkesville (SBCV)	17.54	0.05	16

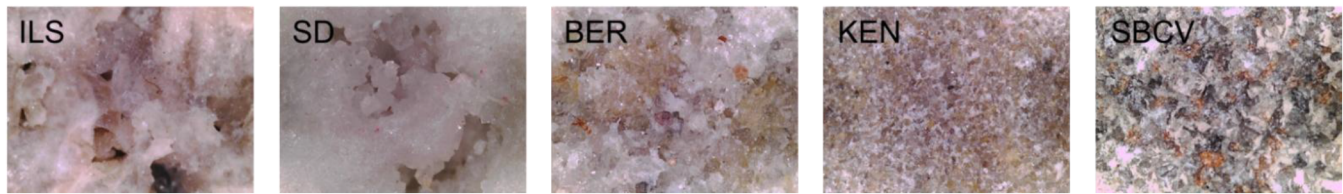


Figure 2—Optical images of the rock surface clearly show pores on surface and also pore surface roughness.

Results and Discussions

The results of the contact angle, IFT, XRD and MICP measurements are presented in this section. The mercury that was used in this study for the contact angle and IFT measurements was in-house cleaned mercury.

Contact angle

The optical tensiometer method was used to measure the static (sessile drop) mercury contact angle. The measurements were performed on a smooth surface of a glass slide and on each of the rock samples in Table 3, one on cleaned sample material with surface roughness preserved (or as received AR) and another on diamond-polished surface (Polished). In each case three measurements were carried out, on three different mercury drops at three locations on the rock surface.

Table 3—Contact angle, variation between the three measurements and MICP intrusion pressure.

	θ_{AR} (deg)	$\theta_{Polished}$ (deg)	$\theta_{Polished} - \theta_{AR}$ (deg)	$std(\theta_{AR})$ (deg)	$std(\theta_{Polished})$ (deg)	P_{intr} (psi)
ILS	137	134	-3	0.1	1.3	2
SD	134	141	7	0.8	0.6	2
BER	137	135	-2	0.4	2.4	5
KEN	139	149	10	7.8	0.6	70
SBCV	140	138	-2	5.4	1.7	120
average	137.4	139.4	2	2.3	6.0	-

Contact angle of mercury on glass. The measured contact angle of mercury on glass surface (smooth surface) as a function of time is shown in Figure 3. The angle was captured for three drops, each for 5

seconds (Figure 3). The average of the three measurements is $136^\circ \pm 1.0^\circ$. A value that is slightly smaller than the standard 140° reported in literature (Table 1).

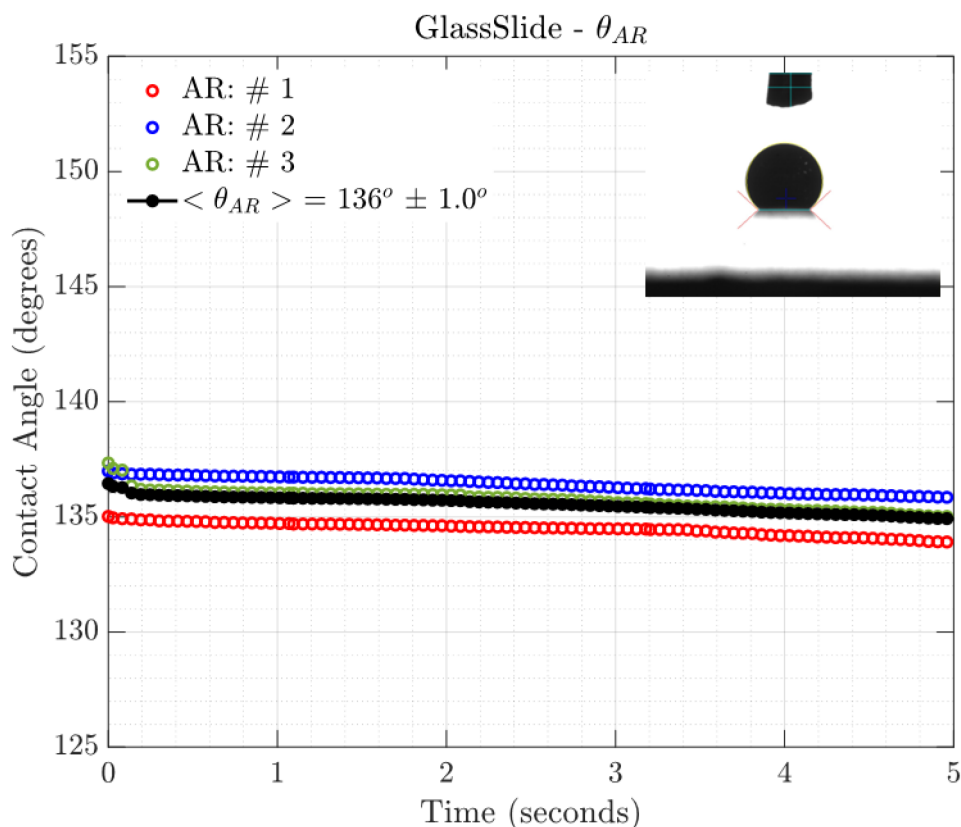


Figure 3—Mercury contact angle as a function of time on glass surface. Black curve represents the average of the three measurements of blue, red, and green circles. Note that differences between the measurements may indicate test repeatability.

Contact angle of mercury on rocks with various lithologies. Contact angle was measured as a function of time on three mercury drops in three different areas on the surface of various rock types, as shown in Figure 4. The measurements on cleaned sample material with surface roughness preserved are shown in the left panels, and on cleaned sample material on a diamond-polished surface in the right panels. To reduce the influence of surface pores and the possibility of drop falling into the pore cavities (Figure 2), the contact angle measurements were performed under static conditions. Therefore, the mercury drop was observed for short period of time; only 5 seconds. In some cases, the three measurements are similar, while in other cases, the intra-sample variability, the difference between the three measurements performed on the same rock sample and the same surface preparation, is observably different. Up to 10% variability was measured in the case of as received KEN sample. For the polished case the discrepancy is smaller, up to 3%, in the case of BER, indicating that polishing may have minimized differences in surface conditions. The intra-sample variability could be due to mineralogy heterogeneity, to the presence of cavities that are crossing the surface, especially when large pores are present, and possibly to the entry pressure.

In the as-received case the contact angle was overall more stable in time, however on the polished surface the contact angle decreased in time for BER and KEN up to about 4%. On the polished surface the resistance to rolling may be reduced and it may be easier for the mercury drop to roll into surface pores (Figure 2), making the drop unstable, and the contact angle would appear to be changing in time.

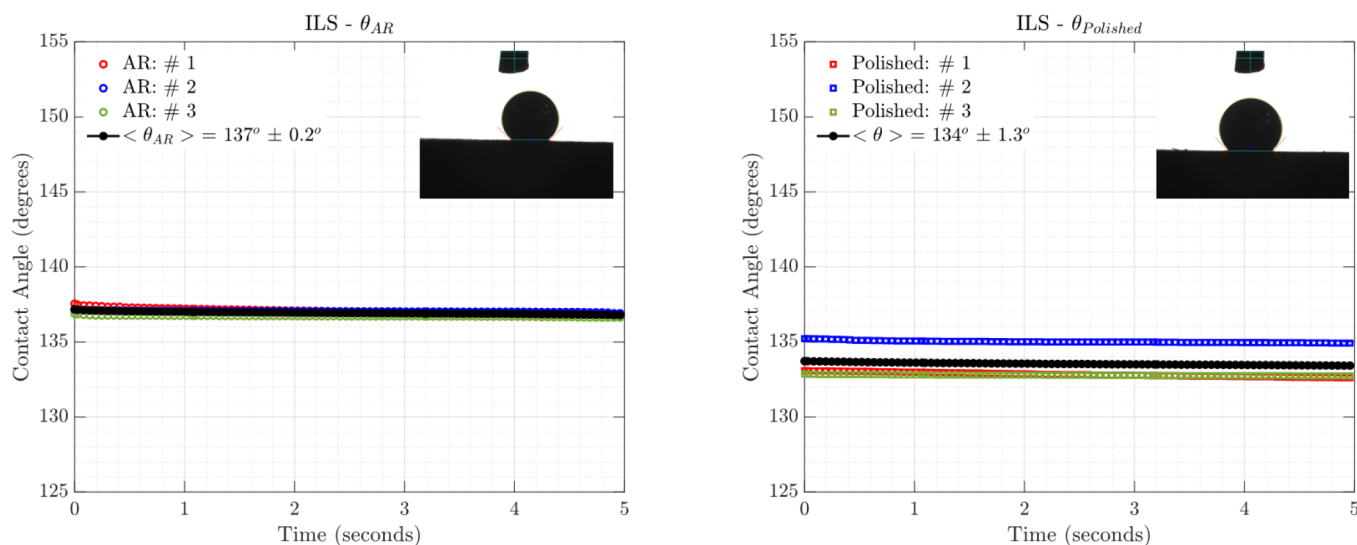


Figure 4a—Measurements of mercury contact angle on as-received (AR) surface (left) vs polished surface (right) - limestone sample ILS. Black curve represents the average of the three measurements. Measurement repeatability on AR surface is much better than on polished surface. Polishing lowered contact angle by about 3° in this case.

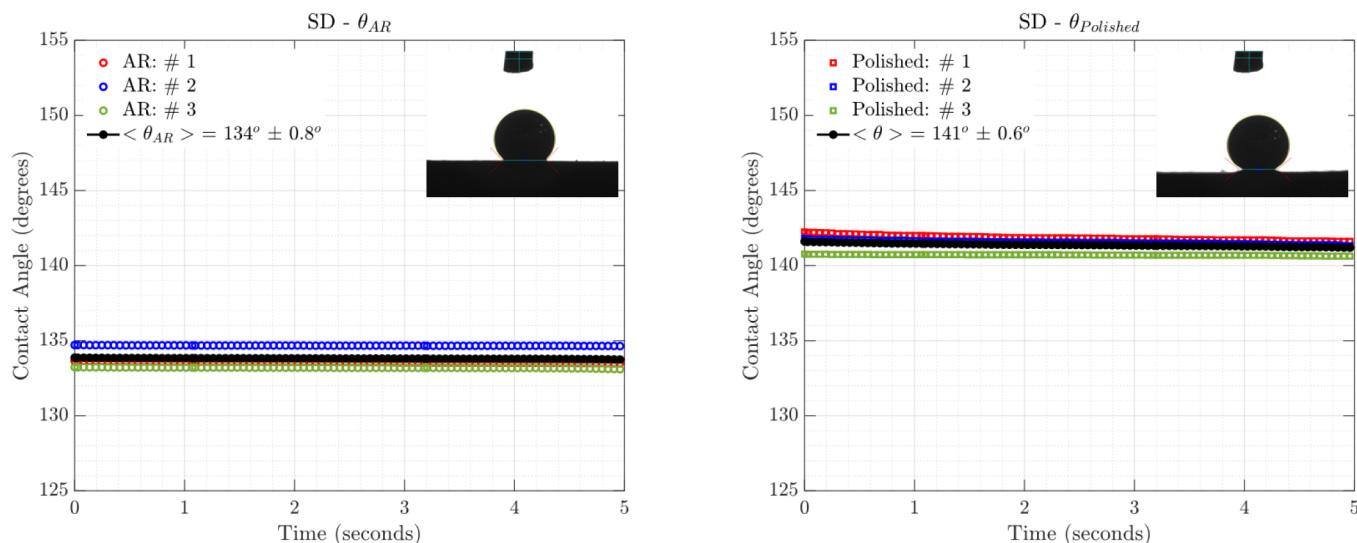


Figure 4b—Measurements of mercury contact angle on as received (AR) surface (left) vs polished surface (right) - Dolostone sample SD. Black curve represents the average of the three measurements (this is an opposite trend for contact obtained for limestone samples in Figure 4a). Measurement repeatability on AR and polished surfaces are similar. Polishing raised contact angle by about 7° in this case.

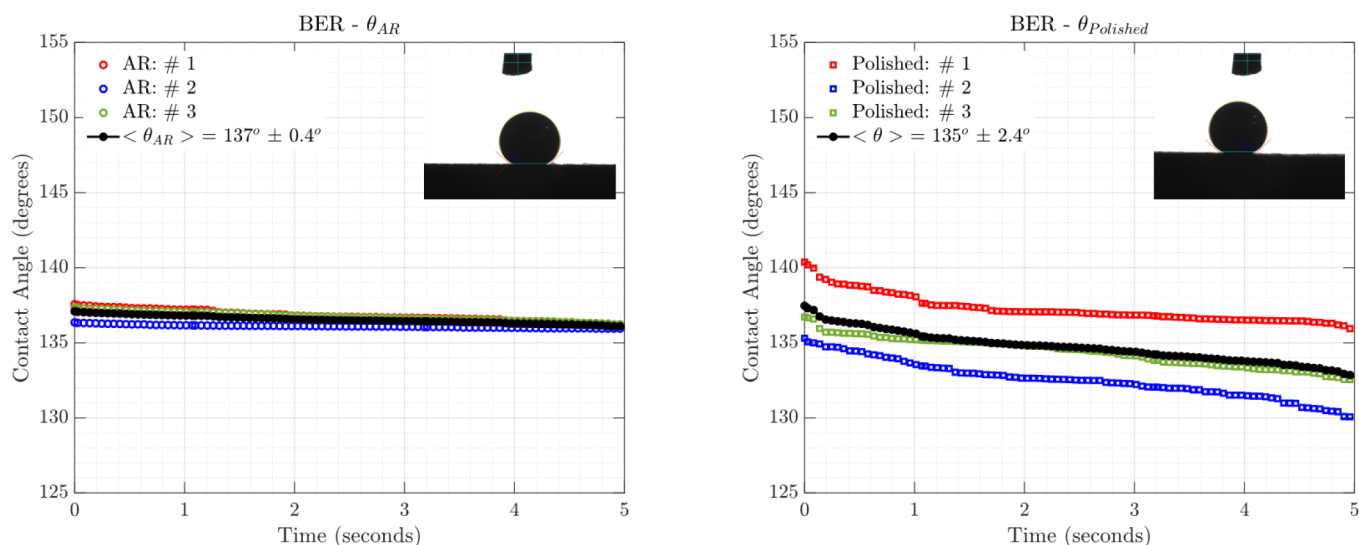


Figure 4c—Measurements of mercury contact angle on as received (AR) surface (left) vs polished surface (right) - coarse grained Berea sandstone sample BER. Black curve represents the average of the three measurements. Measurement repeatability on AR surface is very good, while on polished surface is poor. On average, polishing has little effect on contact angle.

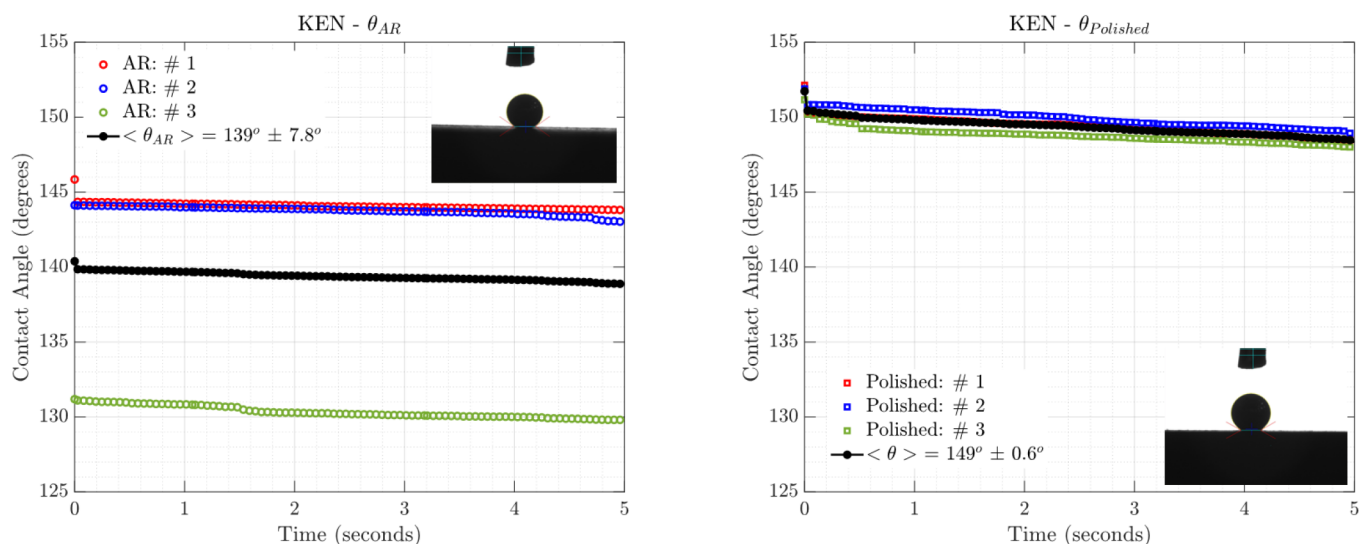


Figure 4d—Measurements of mercury contact angle on as received (AR) surface (left) vs polished surface (right) - fine grained Kentucky sandstone sample KEN. Black curve represents the average of the three measurements. Measurement repeatability on AR surface is very poor, while on polished surfaces is fairly good. On average, polishing raised contact angle by about 10°.

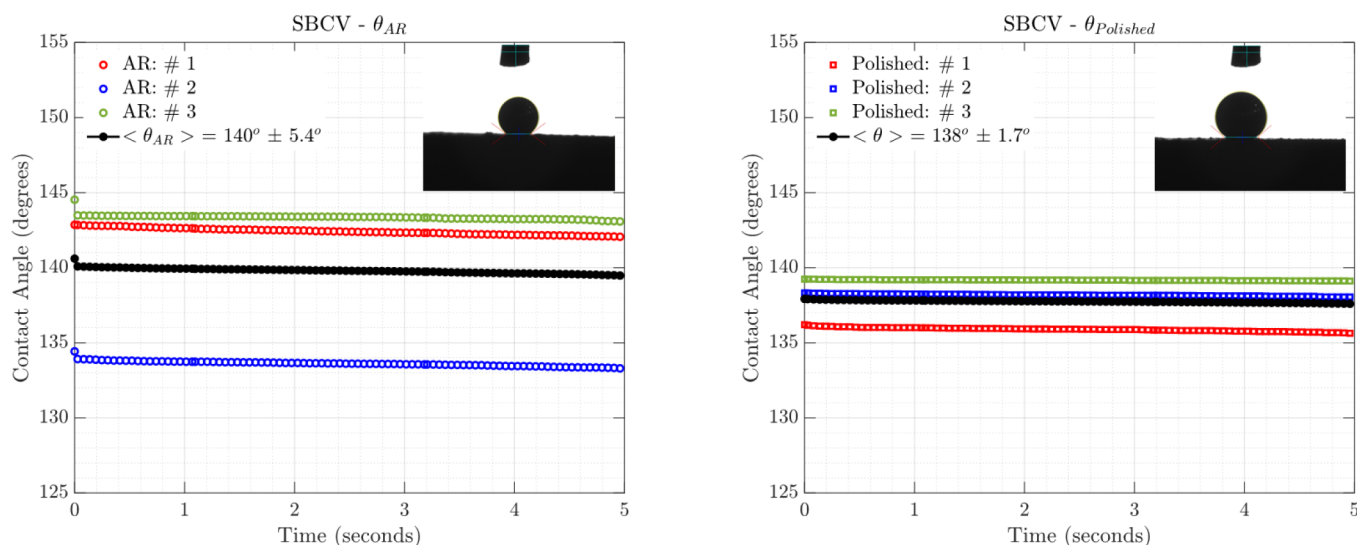


Figure 4e—Measurements of mercury contact angle on as received (AR) surface (left) vs polished surface (right)- clay rich shaly Subclarkesville sandstone sample SBCV. Black curve represents the average of the three measurements. Measurement repeatability on AR surface is very poor, while on polished surfaces is also poor, though it is better than that on AR surface. On average, polishing has little effect on contact angle measurements.

The average of the three measurements of the contact angle of mercury on various lithologies for AR and polished surface is shown in Figure 5, with stars and dots, respectively. The average of all measurements in AR state is $\langle \theta_{AR} \rangle = 137.4^\circ \pm 2.3^\circ$. This translates into an average 4% difference in $\cos$ of the contact angle comparative to the assumed value of 140° , $(\cos(\theta_{AR}) - \cos(140)) / \cos(140) \times 100 \approx 4\%$. The standard deviation is not meaningful for only three measurements; however, we show it in Table 3 with the purpose of highlighting the cases where the three measured contact angle values are not consistent.

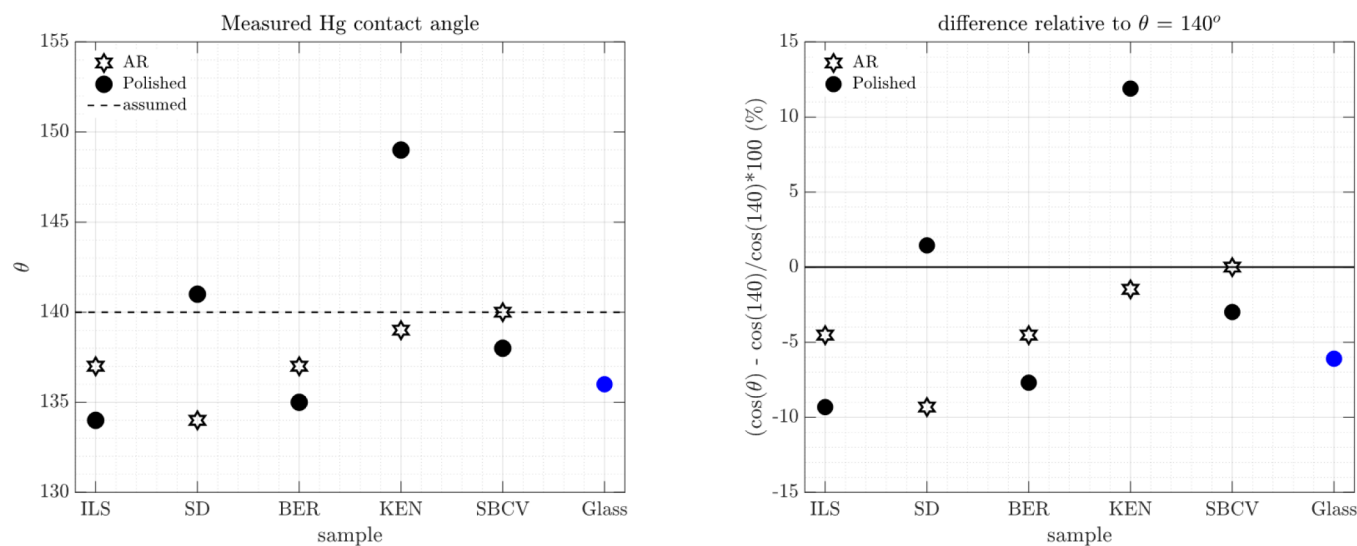


Figure 5—Measured average value of mercury contact angle on various rock surfaces (stars are for as received, dots for polished and the blue dot is for glass slide).

The average of all measurements for the polished surface is closer to the assumed value of 140° , however there is a higher spread compared to the as received case, $\langle \theta_{Polished} \rangle = 139.4^\circ \pm 6.0^\circ$. The effect of surface polishing is not affecting the measured contact angle in a similar, consistent, manner relative to that as received case. Polishing makes the surface very smooth and reduces resistance to rolling which might make

it easier for the mercury drop to fall into surface pores (Figure 2). This effect of polishing could be seen especially in the case of BER and KEN, where the contact angle changes in time.

Possible causes contributing to the differences observed between the contact angle measured on the as-received rough surface, θ_{AR} , and the polished surface, $\theta_{Polished}$, are

- sample heterogeneity,
- presence of surface pores that would make the drop unstable and the measurement uncertain,
- surface roughness, and
- likely measurement repeatability due to difficulties in handling mercury.

The differences observed between the contact angles measured from rough surface and polished surface are smaller than some observed in published results (Sari et al., 2020). They measured the surface roughness effect on the contact angle for oil-brine-rock system, while our case is for mercury-air-rock system. Mercury is unique compared to reservoir fluids, not only because it has a high intrinsic contact angle, but also a high IFT, while air/water and oil/water systems are in the 20-70 dyne/cm range (Mahavadi et al., 2022). The high IFT may help mercury keep its circularity and minimize the influence of changing surface roughness on the contact angle.

XRD measurements

The results of the XRD measurements are shown in Table 4. The weight percent were converted to volume percent using average grain density for each mineral (Al-Hamad et al., 2025).

Table 4—XRD results in % grain volume.

Sample ID	Mineralogy - Volume Percent														
	Quartz	K-Feldspar	Plagioclase	Calcite	(Fe, Mg) Calcite	Siderite	Dolomite	Fe Dolomite / Ankerite	Total Carbonate	Pyrite	Hematite	Halite	Barite	Total Clay	Total
	vol%	vol%	vol%	vol%	vol%	vol%	vol%	vol%	vol%	vol%	vol%	vol%	vol%	vol%	g/cc
ILS	0.9			98.2			0.9		99.1						100
SD	0.9						99.1		99.1						100
BER	84.6	3.8	6.7											4.8	100
KEN	70.3	1.8	19.3											8.6	100
SBCV	73.6			13.9		3.6			17.5	0.0	1.2			7.7	100

Note, blank values indicate the phase was not detected. The grain densities are calculated using the normalized mineral concentrations, good estimates of measured grain densities are shown in Figure 2b of Ma et al., 2002, though they do not consider amorphous material such as bitumen/kerogen or amorphous metal oxides or hydroxides.

Mineralogy and contact angle correlation. The samples richer in mineral types, BER, KEN and SBCV, show a higher difference between the three contact angle measurements (standard deviation larger than 1.5 in Table 3). More minerals mean higher heterogeneity that could explain a higher variation in the measured contact angle. For BER and KEN, the contact angle also decreases with time, and this effect is especially strong for the polished surface. Note that BER and KEN, the only two samples containing K-Feldspar and Plagioclase.

Plots of the entry pressure, P_{intr} , versus quartz and clay percent volume are shown in Figure 6. The entry pressure shows a step change increase with quartz content and the presence of clays seems to indicate an increase in the entry pressure. P_{intr} is less than 5 psi until a clay content threshold of about 5 vol% is reached,

at which point P_{intr} increases up to 120 psi with increasing clay content of up to 8 vol%. For quartz, P_{intr} remains low, less than 5 psi, until a threshold of 70 vol% is reached, at which point P_{intr} becomes independent on the vol%. Note that samples with low quartz and low clay are mostly carbonate. However, more data is required to confirm the trend.

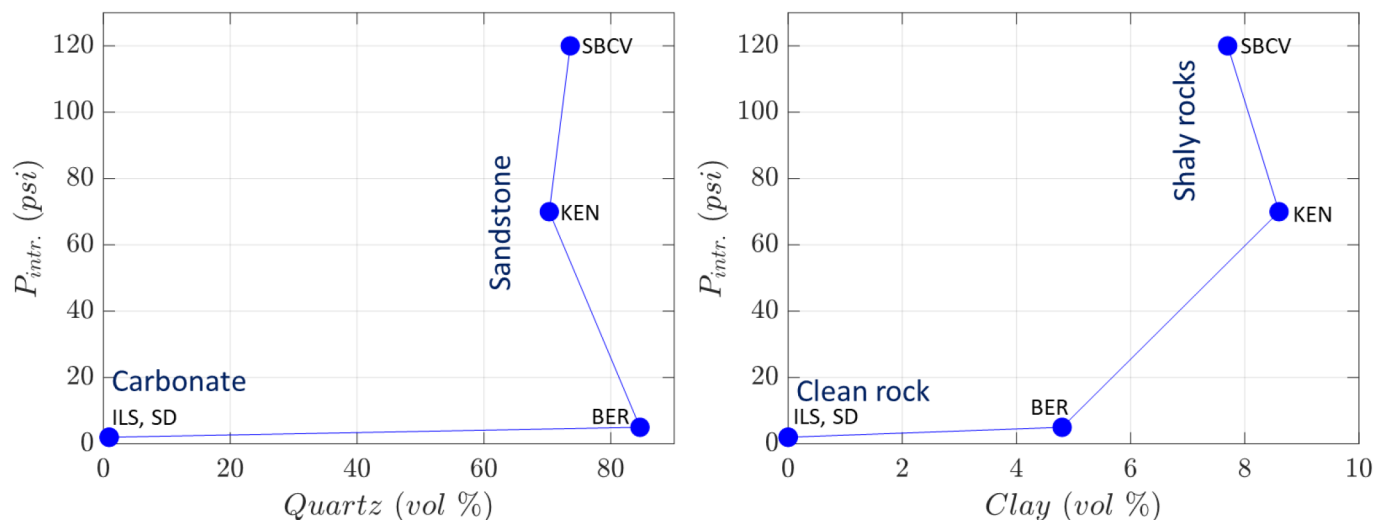


Figure 6—Intrusion pressure vs quartz (left) and clay (right) percent volumes. Entry pressure is inversely proportional to permeability; thus it increases with rock shaliness.

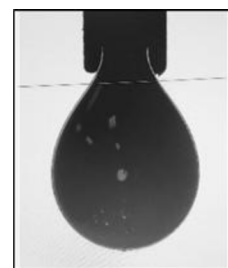
IFT measurements in air

The IFT measurements were determined using the pendant drop methodology by analyzing the hanging drop shape which is a function of IFT and gravity. Typically, IFT measurements are performed prior to setting the drop upon the surface for contact angle measurements. However, because the initial drop sized were not large enough to accurately analyze the drop shape, the pendant drop had to be measured later with an appropriate needle gauge (A 14-gauge 1.6 mm needle was used for the IFT measurement). If the needle gauge is too large the drop falls quickly, while small drop size created by a small needle would have low resolution for accurate image analysis. In our measurements with the 14-gauge 1.6 mm needle, little variation in drop size was observed.

The measured IFT values in air, at 20.5 °C, are shown in Table 5. The mercury IFT average value of six measurements was 417 dyne/cm, 14 % lower than the often-defaulted value of 485 dynes/cm.

Table 5—Mercury IFT measured in air at 20.5 °C, and the difference relative to the often-defaulted value of 485 dynes/cm. The photo illustrates the hanging mercury drop.

	IFT (dyne/cm)	1 - IFT/485
1	391	19%
2	426	12%
3	398	18%
4	433	11%
5	417	14%
6	436	10%
Average	417 ± 4%	14%



Besides the surface it contacts with, the mercury contact angle could be affected by the impurities presenting in the mercury which may result in a lower IFT and contact angle compared to pure mercury. If the rock has metallic mineral, then mercury contamination is likely as mercury is well known for making amalgamation with many metal elements. However, if there are no metals, then mercury would likely remain with a high level of purity even after flowing through rock samples. To verify this, it would be helpful for future work to investigate the effect of mercury cleanliness on the IFT and contact angle comparing quadruple distilled, 99.99% pure, against "dirty" mercury, contaminated from mercury injection with various lithologies. If it is confirmed that the mercury cleanliness does significantly affect the IFT measurements, this would raise a challenging question about what IFT should be used in MICP data interpretation since mercury would be contaminated once entering into rock samples, the less consolidated rocks, the more contamination possible.

Effect of the measured contact angle and IFT on pore throat and body size

Both contact angle and IFT measured values are below the often-defaulted values of 140° and 485 dynes/cm, and as a result both will shift the pore throat and pore body from MICP to smaller sizes (Eq. 1). Figure 7 shows the pore throat size (intrusion), left, and pore body size (extrusion), right, calculated using the default values of 140° and 485 dynes/cm (red), using the measured values for AR surface (green), and the measured values for polished surface (blue). The effect of using the defaulted IFT and contact angle can overestimate the pore throat and pore body size by over 20 %. However, this reduction in pore throat and pore body size from MICP, will not affect the body to throat ratio (Eq. 1), since the effect on both are proportional.

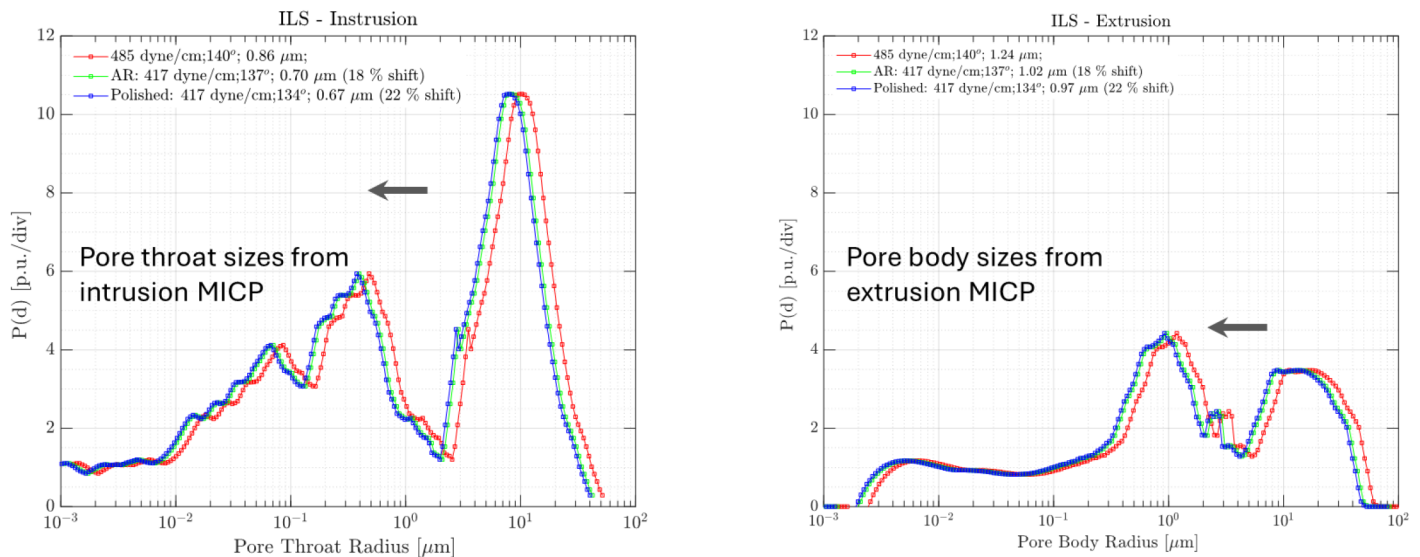


Figure 7a—Pore throat size (intrusion, left) and pore body size (extrusion, right) using the typical 140° and 485 dynes/cm (red), the measured values for AR surface (green), and the measured values for Polished surface (blue) - ILS sample. The arrows indicate the shift towards smaller value for both pore throat and pore body when using the measured values for contact angle and surface tension.

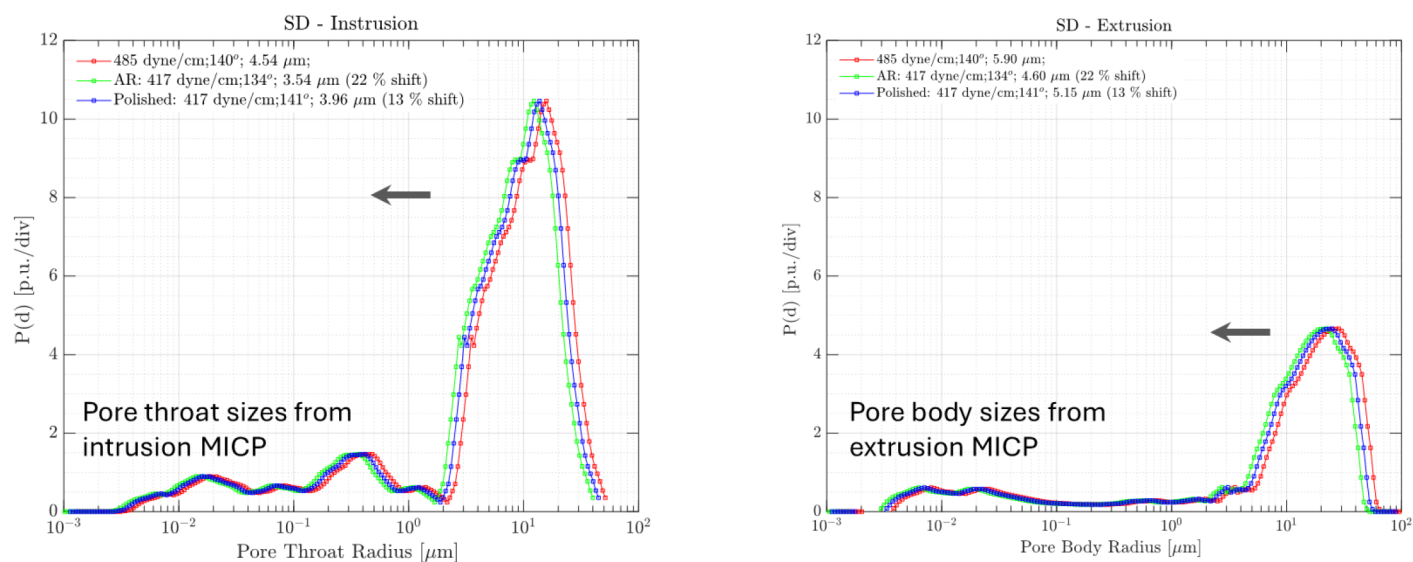


Figure 7b—Pore throat size (intrusion, left) and pore body size (extrusion, right) using the typical 140° and 485 dynes/cm (red), the measured values for AR surface (green), and the measured values for Polished surface (blue) - SD sample. The arrows indicate the shift towards smaller value for both pore throat and pore body when using the measured values for contact angle and surface tension.

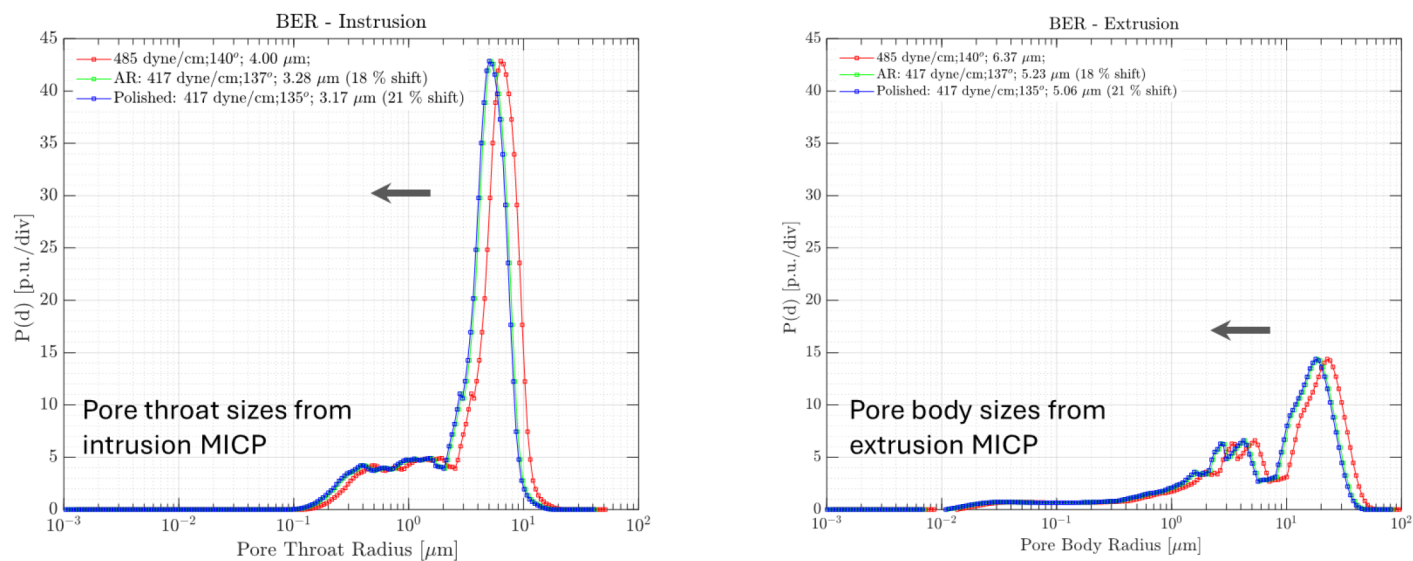


Figure 7c—Pore throat size (intrusion, left) and pore body size (extrusion, right) using the typical 140° and 485 dynes/cm (red), the measured values for AR surface (green), and the measured values for Polished surface (blue) - BER sample. The arrows indicate the shift towards smaller value for both pore throat and pore body when using the measured values for contact angle and surface tension.

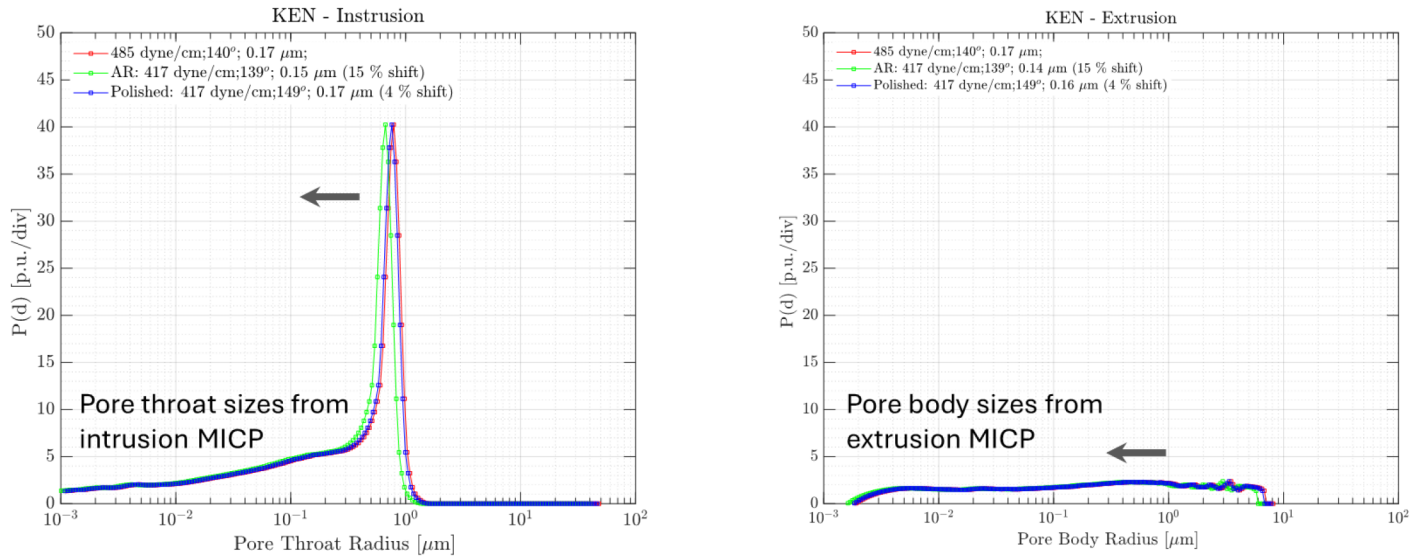


Figure 7d—Pore throat size (intrusion, left) and pore body size (extrusion, right) using the typical 140° and 485 dynes/cm (red), the measured values for AR surface (green), and the measured values for Polished surface (blue) - KEN sample. The arrows indicate the shift towards smaller value for both pore throat and pore body when using the measured values for contact angle and surface tension.

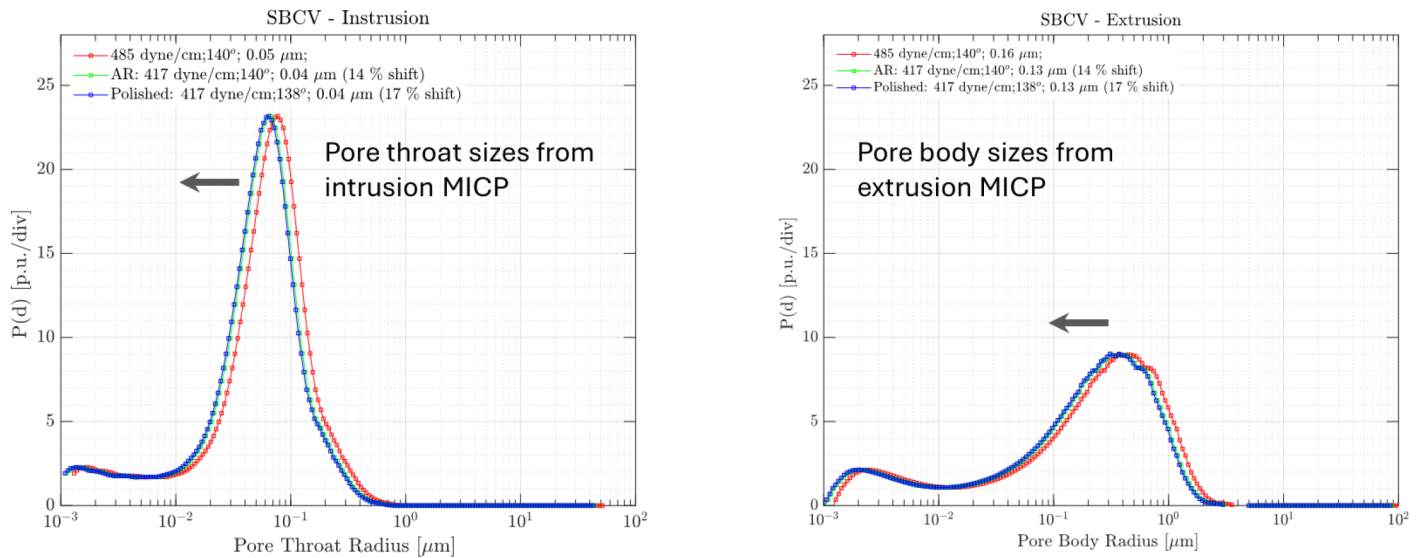


Figure 7e—Pore throat size (intrusion, left) and pore body size (extrusion, right) using the typical 140° and 485 dynes/cm (red), the measured values for AR surface (green), and the measured values for Polished surface (blue) - SBCV sample. The arrows indicate the shift towards smaller value for both pore throat and pore body when using the measured values for contact angle and surface tension.

Results Summary

Below is a summary of the results generated in this study

- The average mercury IFT measured in air was $417 \pm 4\%$ dynes/cm, 14 % smaller than the often-defaulted value of 485 dynes/cm.
- The average mercury contact angle measured on glass was $\theta_{glass} = 136^\circ \pm 1.0^\circ$, 6% smaller than the often-defaulted 140° , consistent with published data.
- The average value for mercury contact angle on as-received, surface roughness preserved rock, was $\langle \theta_{AR} \rangle = 137.4^\circ \pm 2.3^\circ$, slightly less than the often-defaulted value of 140° by $\sim 4\%$. For polished

surface the contact angle values were more scattered, $\langle \theta_{Polished} \rangle = 139.4^\circ \pm 6.0^\circ$. For ILS, BER and SBCV, $\langle \theta_{AR} \rangle > \langle \theta_{Polished} \rangle$, while for SD and KEN $\langle \theta_{AR} \rangle < \langle \theta_{Polished} \rangle$.

- Factors contributing to the contact angle inter-sample variability:
 - Surface roughness
 - Variation in mineral types
 - Pore entry variability; even for polished surfaces there may be variability depending on pore openings. Polishing will make the surface very smooth and reduce resistance to rolling which might make it easier for the mercury drop to fall in the cavities that cross the surface.
- Factors contributing to the contact angle intra-sample variability:
 - Surface heterogeneity - it was observed that the richer the mineralogy, for BEN, KEN and SBCV, the higher the standard deviation; for ILS and SD, almost 100% calcite and dolomite respectively, the standard deviation was lower.
 - Surface roughness

More work needs to be done to quantify the uncertainties, the effect of mineralogy heterogeneity and to investigate more thoroughly what causes the differences between polished and unpolished measurements. Are they mainly due to surface roughness, the dropping site of mercury on the rock surface, the presence of cavities on the surface?

Conclusions

Based on this laboratory study, the following can be concluded:

Due to its unique characteristics of begin heavy with high IFT, measurement of its IFT in air and contact angle with solids are difficult, resulting in scatterness in measurement results.

The scatterness in measured contact angle across the same sample appears to correlate with the heterogeneity in mineralogy of the sample. Specifically, the limestone and dolostone samples show minimal scatterness in contact angle with simpler mineralogy, while the sandstone and clay-rich sand show more scatterness in contact angle with more complex mineralogy.

Results from this study indicate that the values measured for the mercury IFT in air and contact angle on various lithologies are both smaller than the typically defaulted values of 140 degrees and 485 dynes/cm, as summarized in the table below, which result in an average of 17% reduction in pore throat sizes and body sizes derived from MICP intrusion and extrusion, respectively, with little effect on the derived pore body-to-throat ratio.

	σ (dynes/cm)	θ (degree)	$\sigma \cos(\theta)$	$\left(1 - \sigma \cos(\theta) / (\sigma \cos(\theta))_{default}\right)$
Default values	485	140	-371.5	
Measured values	417	138	-309.9	16.6%

It is noted that mercury contact angle for shale source rocks may need further studies. In addition, more thorough investigations on effects of impurities present in the mercury, i.e. rock-solid particles, uncleaned crude oil, deposited salts, etc., on mercury IFT, as well as its contact angle with rock pore surface may also need future studies.

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